UNIVERSITY B D T COLLEGE OF ENGINEERING

DAVANAGERE - 577 004

(A Constituent College of VTU, Belagavi)

SYNOPSIS ON MINI PROJECT (CODE **: BEC586)**

"SMART TRASH BIN USING ULTRASONIC SENSOR"

Submitted in partial fulfilment of the requirement for the degree of

Bachelor of Engineering in Electronics and Communication Engineering

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Table of contents:

1.	Introduction	3
2.	Problem statement	3
3.	Objectives	3
4.	Methodology	4
5.	Components required	5
6.	Advantages and Applications	5-6
7.	Expected outcome	6
8.	References	7

1. Introduction

As the world is in a stage of up gradation, there is one stinking problem we have to deal with. Garbage! In our daily life, we see the pictures of garbage bins being overfull and all the garbage spills out. This leads to the number of diseases as large number of insects and mosquitoes breed on it.

In India, the traditional waste management system is collecting and clearing garbage regularly and periodically by the municipal corporation or competitive authority. This system will be more effective and non-hazardous with the municipal people. When no toxic material will be in contact with them. A society will get its waste dispatched properly only if the trash bins are placed and emptied properly. The working of the project is based on the Arduino platform. The Ultrasonic sensor is connected at the top of trash bin. It is used to detect the level of garbage filled at any instance of time. If the trash bin is full, then LED will show the status of the bin.

2. Problem Statement

Overflowing of trash bins and foul Odors create unhygienic conditions in waste management, as conventional trash bins require frequent manual checking and lack monitoring features. A smart trash bin solves this problem by monitoring waste levels in real time.

3. Objectives

1. To monitor waste levels automatically using sensors.

2. To prevent overflow and maintain cleanliness.

3. To reduce manual checking of trash bins.

4. To give alerts/indications (like LEDs, display, or buzzer) when the bin is full.

5. To support smart city waste management systems.

6. To promote hygiene and reduce health risks.

4. METHODOLOGY

The smart trash bin is designed to be simple and easy to assemble, with readily available components. The main components of the trash bin include an Arduino microcontroller, a servo motor, an ultrasonic sensor, a power source, gas sensor and a led.

The Arduino microcontroller is the brain of the trash bin and is responsible for controlling the servo motor and the ultrasonic sensor. The servo motor is used to open and close the lid of the trash bin, while the ultrasonic sensor is used to detect the presence of objects. The ultrasonic sensor is mounted on the lid of the trash bin and uses high-frequency sound waves to detect objects. When an object is detected, the Arduino microcontroller sends a signal to the servo motor to open the lid, allowing the user to dispose of their waste. Once the waste is deposited, the Arduino microcontroller sends another signal to the servo motor to close the lid, ensuring that the trash bin remains closed and Odor-free. Both works propose a smart garbage alert system for proper waste management using an ultrasonic sensor and Arduino UNO.

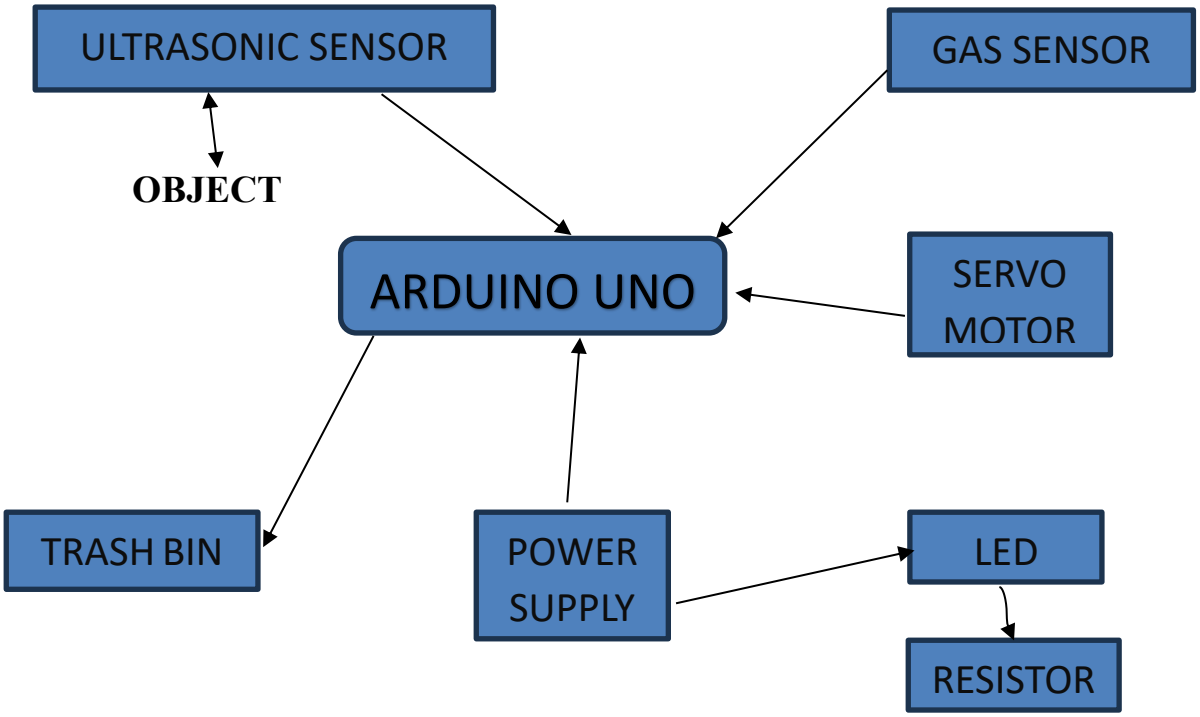

Figure 1: Block diagram of smart trashbin

5. Hardware Components Required

- Arduino UNO:. It operates on 5V and can be powered using USB or an external supply.
- Ultrasonic Sensor: An ultrasonic sensor is an electronic device that measures distance by using sound waves.
- Servo motor: In a smart dustbin, a servo motor is used to automatically open and close the lid.
- Gas Sensor: The gas detector **(MQ-135)** is a sensor used to measure air quality by detecting harmful gases.
- LED: The Light Emitting Diode (LED) is a semiconductor device that emits light when current flows through it.
- Resistor: The **220-ohm** resistor is an electronic.
- Power Supply: A 9V battery is a small rectangular battery.
- Jumper wires: Jumper wires are electrical wires used to make temporary connections.

6. Software tool

The primary software tool required for the smart dustbin project is the Arduino IDE (Integrated Development Environment). It is used to write, edit, compile, and upload programs to the Arduino Uno using a C or C++ based language. The IDE also includes a serial monitor for real-time data observation, which helps in testing and debugging sensors like ultrasonic and gas detectors. Its simple interface and wide library support make it ideal for developing and managing embedded system projects.

7. Expected Outcome

The lid of the dustbin is successfully opening when an object comes in front of the sensor within the 30cm of distance. By using ultrasonic sensors and servo motors, the smart dustbin can automatically open and close its lid, making it more convenient for people to dispose of their waste without having to touch the bin. Also detects the waste level using an ultrasonic sensor and indicates the status through red and green LEDs. It also senses bad odor using a gas sensor and alerts users when the bin needs cleaning.

8. Advantages

- Prevents overflow – Keeps the surroundings clean.
- Hygienic – Reduces bad smell and spread of diseases.
- Efficient waste collection – Alerts when the bin is full.
- Smart city support – Can be linked with IoT systems.
- User-friendly – Easy to use and maintain.
- Cost-effective in long run –Components used are affordable.

9. Applications

- Public Places: Can be installed in parks, bus stands, and railway stations to maintain hygiene by monitoring waste levels and Odors.
- Smart Cities: Useful in smart waste management systems for efficient garbage collection .
- Hospitals and Offices: Helps maintain cleanliness and odor-free environments with automated monitoring.

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